

Customer Churn Analysis for OTT Streaming Platform

Final Project Report

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1. Executive Summary

This project analyzes customer behavior for an OTT streaming platform to identify the factors influencing customer churn, subscription upgrades, revenue generation, and customer satisfaction. The analysis was performed using Excel, SQL Server, and Power BI.

The final dashboard provides insights into customer retention, upgrade opportunities, revenue performance, support experience, and customer risk levels. These findings can help the business improve retention, increase revenue, and make better strategic decisions.

2. Project Overview

The OTT streaming platform wanted to better understand customer behavior and identify opportunities to improve business performance. The project involved analyzing customer data, subscription details, platform usage, support interactions, and churn history.

3. Business Problem

- High customer churn
- Difficulty identifying customers likely to upgrade
- Limited visibility into customer engagement
- Need for actionable business insights

4. Project Objectives

- Analyze customer churn patterns
- Identify customers suitable for upgrade campaigns
- Evaluate customer engagement and support behavior
- Compare revenue generated by different subscription plans
- Build interactive dashboards for decision-making
- Provide business recommendations

5. Dataset Description

Table	Description
Customer	Customer demographic information
Subscription	Subscription plan details
Usage	Platform usage and engagement
Support	Customer support interactions
Churn	Revenue, upgrade history, and churn status

6. BRD & FRD Summary

The business required a reporting solution capable of **tracking churn, upgrades, revenue, customer engagement, support performance, and customer risk levels**. The functional requirements were implemented through **SQL analysis** and **Power BI dashboards**.

7. Data Preparation (Excel)

Data cleaning included **duplicate checks, missing value identification, data type correction, formatting**, and basic **exploratory analysis** using **Pivot Tables** and **charts**.

8. SQL Analysis

SQL Server was used to calculate **churn rate, upgrade acceptance rate, revenue trends, top customers, tenure analysis, monthly churn, and customer risk categorization**.

9. Power BI Dashboard

Page 1 — Executive Overview

Churn Rate: 49.03%

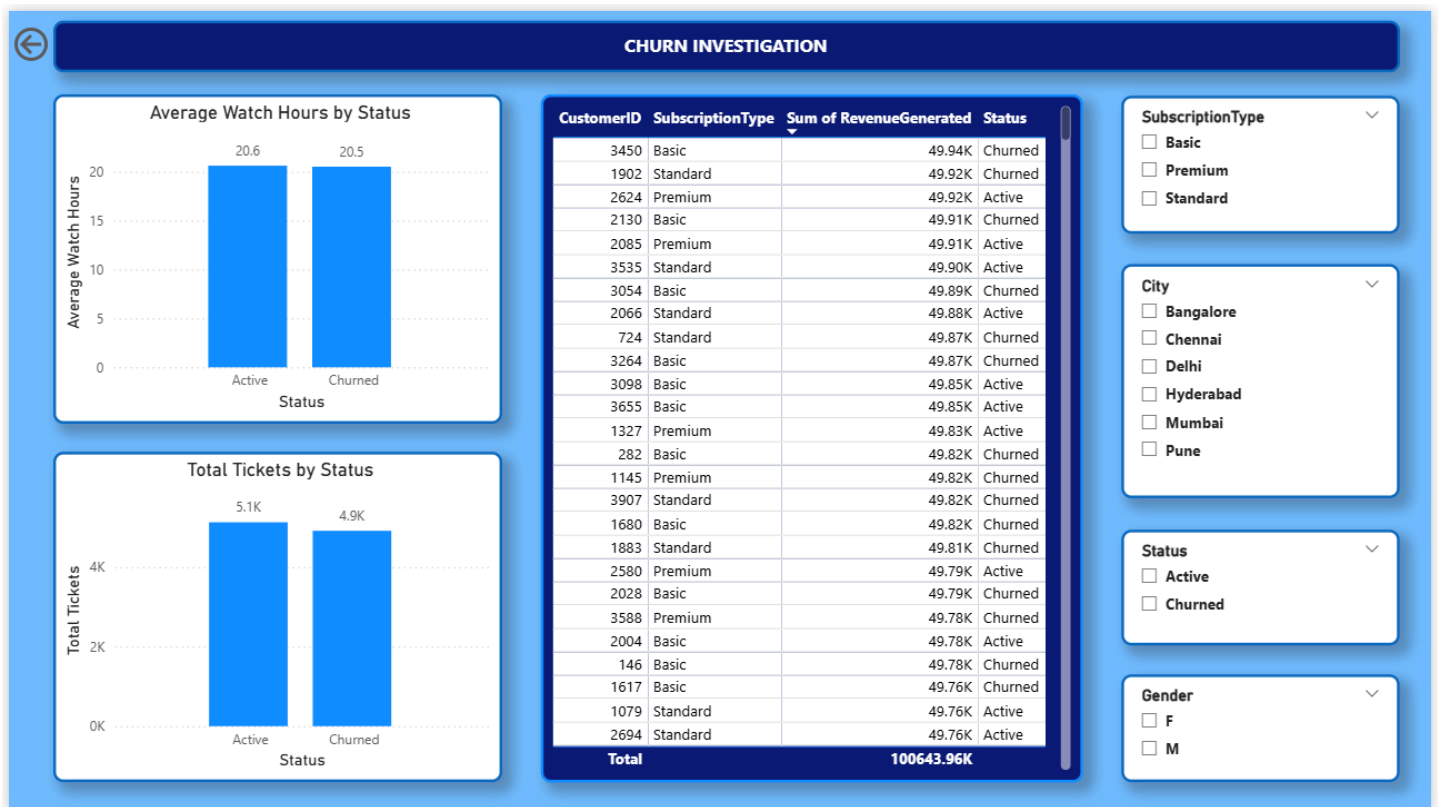
Upgrade Acceptance Rate: 51.93%

Total Revenue: 101M



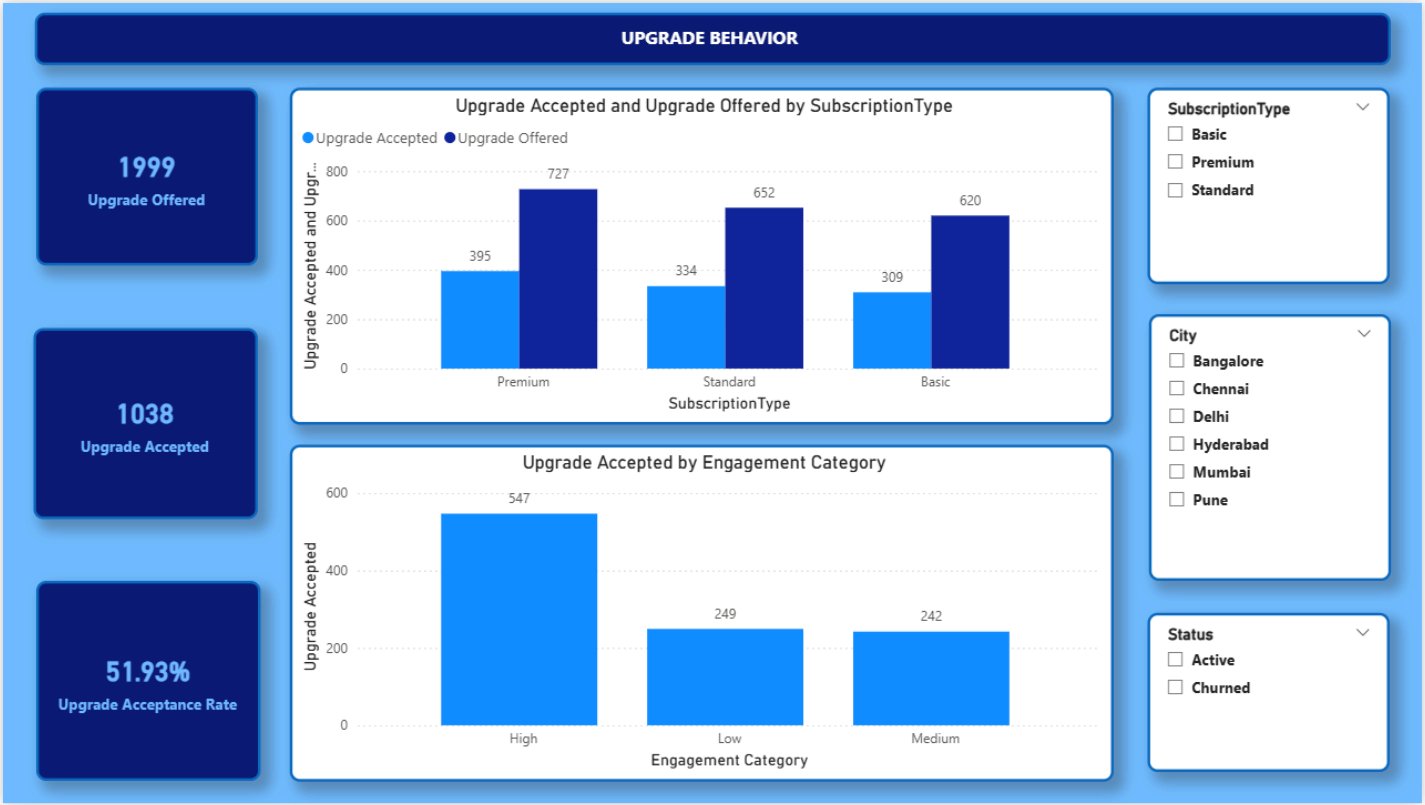
Short Insight:

- The platform generated **101M** in total revenue.
- The **49.03%** churn rate indicates a significant retention challenge.
- More than half of the customers offered an upgrade accepted it (**51.93%**).



Short Insight:

- Customers with **lower engagement** are more likely to **churn**.
- **Support interactions** are **higher** among **churned customers**.
- Customer-level **drill-through** helps **identify specific churn patterns**.



Short Insight:

- **Highly engaged customers show the highest upgrade acceptance.**
- Upgrade campaigns are performing well with a **51.93%** acceptance rate.
- **Basic and Standard** users represent **strong upgrade opportunities**.



Short Insight:

- Premium Subscription generated the highest revenue (35M).
- Hyderabad is the highest revenue-generating city.
- Top customers contribute a significant share of total revenue.



Short Insight:

- **163** churned customers had support interactions before leaving
- Customers have a **moderate level of satisfaction (3.77/5)**, indicating opportunities to improve the overall user experience and customer support to increase retention. (**Average Rating = Sum of all ratings / Number of Rating**)
- Risk analysis helps identify customers needing proactive retention efforts.

10. Business Insights

- **Why are customers churning?**

Customers with low engagement and higher support interactions are more likely to churn. The overall churn rate of 49.03% highlights the need for stronger retention strategies.

- **Which customers should be targeted for upgrade?**

Highly engaged customers should be prioritized for upgrade campaigns because they show the highest likelihood of accepting upgrade offers.

- **What type of users are risky?**

Customers with **low watch hours**, **frequent support tickets**, and **moderate satisfaction** levels represent the **highest risk** group among active users.

Customers with **low watch hours**, **multiple support tickets**, and **moderate customer ratings** should be considered Medium Risk and monitored closely.

These users should be targeted with retention campaigns before they decide to leave the platform.

- **Which subscription is most profitable?**

Premium Subscription is the most **profitable plan** with approximately **35M** in revenue.

Standard Subscription generated **34M**, while **Basic Subscription** generated **32M**.

Increasing the number of **Premium subscribers** can significantly improve overall business revenue.

11. Recommendations

- Improve **customer engagement** through personalized content recommendations.
- **Expand targeted upgrade campaigns** for **highly engaged** users.
- Improve **support response time** and **issue resolution**.
- Monitor **medium-risk customers** proactively.
- **Promote Premium subscriptions** to **increase overall revenue**.
- **Replicate successful strategies** from Hyderabad in other cities.

12. Conclusion

This project successfully transformed raw customer data into meaningful business insights using Excel, SQL, and Power BI. The dashboards provide a clear understanding of customer behavior, churn patterns, upgrade opportunities, revenue trends, and support performance. The recommendations can help improve customer retention, increase revenue, and support better business decisions.

14. Skills Demonstrated

Business Analysis

- Requirement Gathering
- BRD Preparation
- FRD Preparation
- Business Analysis

Excel

- Data Cleaning
- Pivot Tables
- Charts
- Data Validation

SQL

- Joins
- CTE
- Window Functions
- CASE Statements
- Aggregate Functions

Power BI

- Data Modeling
- DAX
- Interactive Dashboards
- Drill-through
- Tooltips
- KPI Cards
- Data Visualization